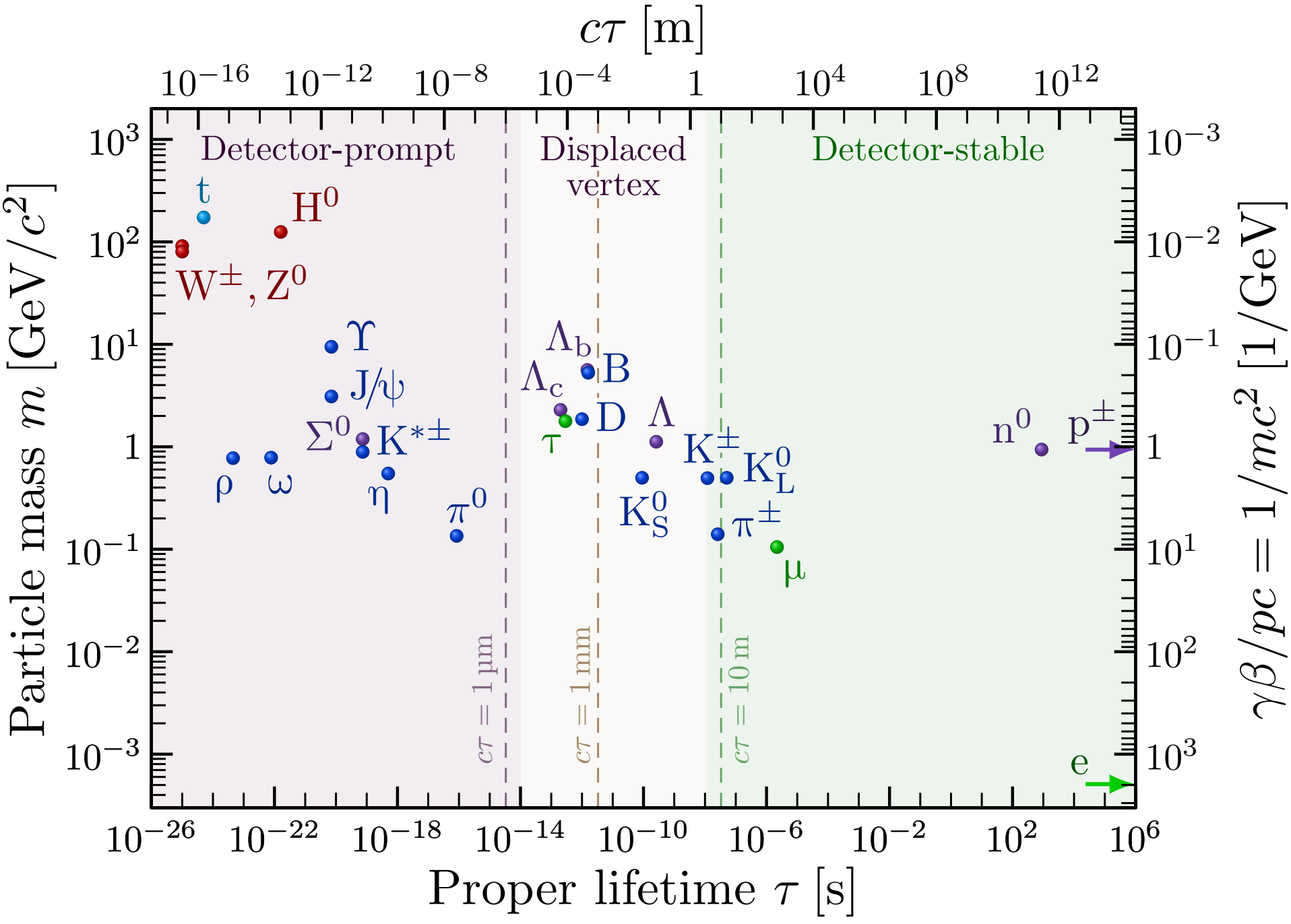

$c\tau$ [m]
10^{-16} 10^{-12} 10^{-8} 10^{-4} 1 10^{4} 10^{8} 10^{12}
Detector-prompt
Displaced vertex
Detector-stable
t
H^0
W$^\pm$, Z^0
Υ
J/ψ
Σ^0
K$^{*\pm}$
ρ
ω
η
π^0
Λ_b
B
Λ_c
D
τ
Λ
K$^\pm$
K^0_L
K^0_S
$\pi^\pm$
μ
n^0
p$^\pm$
e
$c\tau = 1\,\mu$m
$c\tau = 1\,$mm
$c\tau = 10\,$m
Particle mass m [GeV/c^2]
10^{3} 10^{2} 10^{1} 1 10^{-1} 10^{-2} 10^{-3}
$\gamma\beta/pc = 1/mc^2$ [1/GeV]
10^{-3} 10^{-2} 10^{-1} 1 10^{1} 10^{2} 10^{3}
10^{-26} 10^{-22} 10^{-18} 10^{-14} 10^{-10} 10^{-6} 10^{-2} 10^{2} 10^{6}
Proper lifetime τ [s]